Beyond Culture as Facts: Surfacing Systemic Challenges and Community-Defined Knowledge for Generative AI Evaluation

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Abstract

As generative artificial intelligence becomes a more common source of information, questions remain about how well these systems understand culture at the local level. This study examines how systemic challenges are incorporated into community-defined cultural knowledge. Using qualitative data from community workshops across the US, the study analyzes participant-generated data to dive into how participants view and define local cultural knowledge. Through a series of in-person exercises participants generated taxonomies and specific cultural topics that reflect their lived experience in their locations. The analysis identified four themes ranging from everyday life to community history and identity. The findings demonstrate that participants understood cultural knowledge as extending beyond traditional cultural markers. Social, environmental, and economic concerns are also culturally relevant because they shaped everyday experiences and relationships with local places and institutions. The study suggests that evaluating cultural understanding in generative AI requires more than assessing factual accuracy. Community-defined approaches can provide important local context for evaluating how AI systems understand and represent communities. This paper ends with a set of recommendations for evaluating and measuring systemic issues as part of cultural knowledge.

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CCS Concepts: • **Human-centered computing** → **Ethnographic studies**; *Human computer interaction (HCI)*; • **Social and professional topics** → **Cultural characteristics**; *User characteristics.*

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1 Introduction

Artificial Intelligence (AI) has rapidly entered human information seeking, creation, and interpretation behaviors. Large language (LLMs) and other generative models demonstrate extraordinary proficiencies in answering questions on a wide range of subjects. However, responding to questions about a community is not equivalent to understanding that community. This limitation is especially evident in attempts to describe culture, history, and lived experiences requiring deeper insights than general knowledge accessible to all.

1.1 Community and Cultural Knowledge

Communities often acquire knowledge by living in a place and engaging in activities that shape their culture. This may include food, music, institutions, and history, as well as the names of local places, the boundaries of neighborhoods, unique experiences, and circumstances faced by residents [14]. Some of these characteristics are documented, and others are only known to those who have lived in a community and can identify with a place and its people. Consequently, an AI may provide correct information on a community but fail to grasp essential elements that contribute to a community's understanding of itself.

For this study, cultural knowledge is defined as the set of local histories, traditions, institutions, experiences, and conditions that are identified by community members as contributing to their communal understanding of the communities where they live. This study considers community members as active agents in culture definition, and their awareness of what constitutes cultural knowledge may extend beyond the common understanding of culture. The lived experience and sociostructural conditions which ground local knowledge can intersect with other elements of culture to become a part of it.

1.2 Representation Evaluation And Cultural Heuristics - REACH

The importance of understanding culture extends beyond satisfying intellectual curiosity about the world. The Representation Evaluation And Cultural Heuristics (REACH) AI Consortium is a multi-stakeholder partnership between Google and 12 Historically Black Colleges and Universities. REACH is an initiative that seeks to develop conceptual and methodological approaches to assessing responses to culturally probing questions generated by large multimodal language models from the perspective of hyperlocal communities. The broader REACH project implements community-informed automated evaluation strategies to interrogate questions related to sensitive topics and historical events to develop more complete conceptualizations of cultural knowledge.

Part of the REACH effort is to include community perspectives by design in the evaluation process. As such, in-person workshops were held in 6 different locations across the United States: Washington, D.C., Los Angeles, California, Baltimore, Maryland, Tuskegee, Alabama, Chicago, Illinois, and Taos, New Mexico. The objectives of the REACH workshops were to identify topics of interest to each community, define community-specific concerns, and formulate questions to ask an AI. The data collected from the REACH workshops provide insights into the topics, domains, and issues that residents of these communities consider essential elements of cultural knowledge. This research explores the themes emerging from the analysis of qualitative data obtained from the workshops to explore the relationship between community-defined cultural knowledge and the influence of local challenges on community definitions of culture. Our work asks the broader question, *how do communities define local culture?* However, this paper focuses on the more specific research question: *How are systemic challenges incorporated into community-defined cultural knowledge?* and how can this be measured for the purposes of evaluating Generative AI outputs?

2 Related Work and Background

We situate our work in the interdisciplinary field of AI, participatory design, and cultural studies. As AI continues to expand in everyday use, understanding what LLMs "know" about the culture of a particular geogrpahy or social group is crucial. Generative models can rapidly output facts about countries and cities, but a critical distinction exists between cultural information akin to encyclopedic or trivia-like knowledge and a contextualized understanding of local communities.

2.1 Community-Defined Taxonomies and Participation

Determining what to evaluate from a local cultural perspective is inherently challenging. Relying solely on external experts risks reducing lived experiences to rigid, top-down categories. Instead, we ground our approach in emergent classification and folksonomy, where categories develop organically through how community members organize and describe information [24]. No classification fully captures a place, but local residents are uniquely positioned to identify what feels authentic, relevant, or misaligned with their experiences.

This bottom-up framework aligns with scholarship advocating for meaningful, non-tokenistic community participation in AI development and evaluation. True participation must go beyond treating communities as mere data sources [4], and cannot be a superficial design fix applied after fundamental system decisions are made [21]. Grounded in design justice frameworks, we emphasize the knowledge of communities most affected by technological systems [8], shifting the paradigm from asking communities to respond to predetermined criteria to including them in defining evaluation priorities.

This is of particular importance in how researchers operationalize the categories they measure. Such categories are often proposed by developers who do not sufficiently ground them in relevant disciplinary literature [5]. As such, evaluations risk overlooking the histories, local institutions, and nuanced concerns that residents prioritize, thereby reproducing reductive, outside representations.

2.2 Cultural Knowledge, Literacy, and Local Alignment

Defining and representing culture remains a systemic challenge in generative AI. A review of over 90 studies reveals that researchers rarely define "culture" explicitly, relying instead on static proxies like nationality, demographic variables, or predefined values [1, 15], whereas culture is inherently dynamic and situated [17]. Furthermore, training data disproportionately represents populations with high digital production capabilities, skewing model exposure [12]. Because LLMs actively participate in circulating cultural meaning rather than retrieving neutral facts [11], this digital divide has downstream representational consequences.

Recent audits reveal a stark gap between knowing cultural facts and understanding cultural context. LLMs demonstrate

highly uneven performance in cultural literacy, culturally specific knowledge, and cultural commonsense across different languages and geographies [7, 20, 22]. Even when models possess meta-cultural competence—the theoretical capability to adapt to different cultural settings [10, 18]—they often struggle to determine when specific local knowledge matters.

This divergence is acute at the local level. While language is frequently used as a proxy for culture, low-resource or localized cultural contexts are consistently poorly represented [13, 25]. Poor representation is not always due to an absolute absence of data; rather, models regularly prioritize generalized national data over granular local facts. Geographically, models have been shown to predict national perception data accurately while performing poorly at city and neighborhood levels, rendering them "nationally representative, locally misaligned" [6].

2.3 Cultural Flattening and Gaps in Hyperlocal Representation

Inadequate local representation leads directly to cultural flattening effect. Model outputs may not be entirely false, but rely on superficial cultural markers, stereotyping, erasure, exoticism, or misappropriation [9, 19]. These systems generalize or place recognizable facts into the wrong contexts, erasing the distinct histories, landmarks, and subcultures that differentiate neighborhoods within the same city or region.

To counter this, researchers must address who possesses the cultural authority to define a community's knowledge. Existing evaluation benchmarks treat culture as uniform, ignoring regional or generational variances [1, 2, 16], which explains why modeled responses often correlate weakly with actual national cultural values [3]. Valuable community knowledge is frequently preserved through oral traditions, folklore, and local institutions rather than online text [26].

Two critical gaps emerge in current literature:

- **Hyperlocal Erasure:** Quantitative benchmarks focus heavily on cross-national or cross-linguistic variations, overlooking hyperlocal knowledge at the city, neighborhood, or institutional level [6, 9].
- **Lack of Contextualization:** Current evaluations miss what community members themselves deem essential to their identity [23]. This includes not only traditions and food, but also how residents navigate local systemic challenges—such as housing inequality, policing, and economic development. At the community level, these structural conditions carry deep cultural and historical meaning.

Our study addresses these gaps by evaluating cultural AI performance through community-identified priorities, exploring how local systemic realities shape community knowledge.

Table 1. Total Participants by Location

Location	Total Participants
Washington, D.C.	25
Los Angeles, California	56
Baltimore, Maryland	38
Tuskegee, Alabama	32
Chicago, Illinois	55
Taos, New Mexico	25
Total	**181**

3 Methods

This study used a qualitative approach to examine how community members define cultural knowledge and evaluate representations of their communities. The data were collected through REACH community workshops conducted in six locations: Washington, D.C.; Los Angeles; Baltimore; Tuskegee; Chicago; and Taos. Rather than treating culture as a predetermined set of categories, the workshops provided opportunities for participants to identify topics, experiences, places, institutions and concerns they considered important to understanding their communities through group exercises and discussions.

3.1 Workshop Recruitment and Exercises

Workshops were held from February to June 2026 in each of the six locations. Recruitment was performed by our third-party collaborator during this period in the weeks leading up to the workshops. Workshops recruitment focused on length of time having lived in each specific location with a requirement of at least 5+ years. We aimed to have representation across different racial and ethnic groups as well as gender as much as possible across each workshop, see Table 1 for the number of participants.

Participants were organized into groups around different areas of expertise, namely Education and Learning, Media and Entertainment, Health and Wellbeing, Public Discourse and Communication. Generally, participants were divided into groups of 5-6 people. We held 3 different group and paired brainstorming exercises to help elicit cultural knowledge they found relevant for AI systems to be aware of. The exercises were defined as follows:

1. **Identify Locally Relevant Topics**: In their groups, participants were first asked to channel a cultural ambassador, they were tasked with brainstorming top of mind topics about their neighborhood, what makes their culture distinct, and what do people from outside the area get wrong. For this task, participants were asked to note each individual topic on a post it for use in subsequent exercises.
2. **Organize the Information/Information Mapping**: Participants were next asked to organize or map their topics into broader categories that reflect their mental

Table 2. Topic Boards and Cards by Location

Location	Topic Boards	Topic Cards
Washington, D.C.	6	13
Baltimore, Maryland	6	37
Los Angeles, California	8	42
Tuskegee, Alabama	6	32
Chicago, Illinois	6	30
Taos, New Mexico	7	22
Total	**39**	**176**

map of how they view the information they produced. Participants clustered their post-its on white boards, looking for patterns and analyzing their organization.

3. **Topic Stack cards**: Last, in pairs or smaller groups of 3, participants were asked to go deeper on a single topic area. They were asked to fill out a topic stack card with an query or prompt related to one of their post-its. For this task, participants were asked to consider what about this topic may stump or challenge an AI with respect to their local culture.

3.2 Qualitative Coding and Thematic Analysis

The topic boards and topic stack cards produced by each participant group were treated as distinct research artifacts for further analysis. Using AI assistance, these sources were turned into digital copies as data for how participants organized and described community knowledge. During these workshops we collected a total of 39 topic boards, and 176 topic cards across all workshops. Table 2 summarizes what came out of each individual workshop. Figure 1 shows examples of a topic board and a topic card respectively.

The qualitative analysis used an AI-assisted thematic analysis followed by manual review and interpretation. Participant-generated data were first analyzed using Google Gemini to identify recurring patterns and potential themes across the dataset. Gemini was used as an analytical aid to surface recurring concepts and organize patterns within the participant responses rather than as the final decision-maker in the thematic analysis. Following the AI-assisted analysis, the researchers manually reviewed the underlying data associated with the identified themes. This process involved examining participant-generated materials across sections of the dataset, evaluating whether the patterns identified through the AI-assisted analysis were supported by the original data, and identifying specific examples that illustrated those patterns. The researchers then used these examples to interpret and refine the themes based on the context in which participants described their communities and evaluated AI-generated representations. This process resulted in four overarching themes: (1) systemic challenges become part of the conditions of everyday life; (2) systemic challenges are understood through place and local context; (3) institutions connect systemic challenges to community history and identity; and (4) communities distinguish between acknowledging systemic challenges and being defined by them. The final themes and their interpretation reflect researcher judgment based on manual examination of the participant data, with Gemini serving as a tool to assist with the initial identification and organization of recurring patterns.

4 Findings

Participants folded systemic challenges straight into what counted as cultural knowledge, linking them to the ordinary experience of living in their communities. Housing, transportation, health care, policing, environmental problems, and economic conditions showed up next to the more expected markers—food, music, celebrations, history. These issues weren't treated as separate policy topics. People saw them as a necessary part of cultural knowledge to understand how residents experience and move through their neighborhoods. We identified 4 themes at the intersection of systemic issues and participants' perspectives on their culture.

4.1 Theme 1: Systemic challenges become part of the conditions of everyday life

The Los Angeles material makes this theme especially clear: One group described the city as "diverse but silo-ed," marked by "geographical + cultural boundaries." On the same board they wrote "lack of transport. not walkable," then listed neighborhood music and art scenes, local theater, food communities, social life, unreliable public transit, and traffic. Another group talked about "many density nodes," the sharp differences between places like Beverly Hills and Echo Park, "Distance/neighborhoods," and "Life based around traffic." Cultural diversity, in their view, couldn't be separated from the physical and structural conditions that shape how people get around.

Housing and displacement worked the same way. When one Los Angeles participant evaluated an AI response about changes tied to major development and Olympic preparation, they pushed back hard: the response focused too much on future anxiety and not enough on what was already happening. "It's more than just anxiety about what could happen. It already is happening and causing displacement." They also wanted more attention to residents' health and well-being and to historical context. Displacement wasn't framed as an abstract policy problem; it was an ongoing condition that shapes how communities are lived and whether they hold together.

The Topic Stack cards Los Angeles participants wrote reinforce this. They didn't just ask questions about cultural characteristics. They asked about the conditions shaping everyday experience. One person asked, "What is the homelessness situation in Los Angeles?" and then criticized the

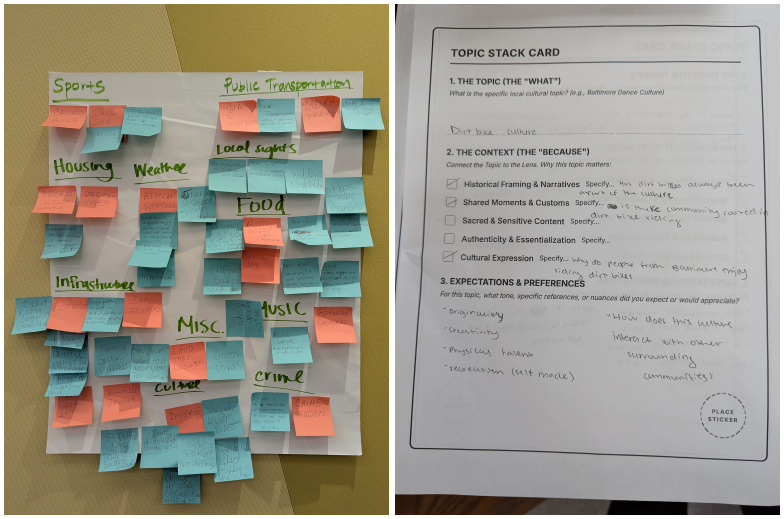


Figure 1. Left: An example Topic Board from the Chicago, Illinois in-person Workshop in April 2026. Topic post-its show areas like sports, housing, infrastructure, public transportation, food, etc. Right: An example Topic card from the Baltimore, Maryland in-person Workshop in March 2026, specifying dirt bikes as its own culture impacting surrounding communities.

answer for ignoring working people and students who are unhoused, housing costs, and the role of state and local government. Another asked how the creation of UCLA "impacted the socio-economic culture of Westwood and affected the people who lived in Westwood before 1929," tying institutional growth directly to cultural change and earlier residents' lives. A Sherman Oaks participant asked about the January 2025 fires in terms of "how it affected your ability to work and commute to and from West LA." These questions treat knowledge about Los Angeles as incomplete if it leaves out the structural conditions that affect housing, mobility, work, and people's relationship to place.

4.2 Theme 2: Systemic challenges are understood through place and local context

Participants also connected their systemic challenges to specific neighborhoods, institutions and other locally meaningful places. This demonstrates how the meaning of a systemic challenge can change depending on where and how it is experienced.

Housing, transportation, policing, environmental concerns and access to resources may affect an entire city, but community members experience those issues locally. Neighborhood boundaries, histories and local conditions provide context for understanding why particular challenges matter. In Northern New Mexico, participants incorporated water access into their understanding of local culture. Discussions of acequias, senior water rights, and the Abeyta Settlement situated water not simply as an environmental or infrastructure issue, but as part of the history and continued life of the community.

Baltimore participants similarly connected systemic issues to locally specific cultural practices. Many boards mentioned dirt bike culture, this was further evidence in the Topic Stack that highlighted "the tension between dirt bike riders and city officials." The topic places a broader tension involving public space, policing, and city authority within the context of a cultural practice associated with Baltimore.

Participants in Taos also framed systemic challenges through specific local institutions and places. Workshop boards identified the region's medical desert as a community concern, while the Topic Stack connected health-care access to the potential closure of Holy Cross Hospital. Participants also raised concerns about Taos Ski Valley related to outside ownership, development and economic change. Rather than discussing health-care access or economic development as abstract issues, participants situated these challenges within institutions and places that were part of their understanding of the local community.

4.3 Theme 3: Institutions connect systemic challenges to community history and identity

Local institutions emerged as important points where systemic challenges, history and community identity intersect. Participants identified schools, universities, government agencies, cultural institutions and other organizations as important to understanding their communities. These institutions could represent sources of pride and community identity while also being connected to concerns about resources, access, inequality or other structural challenges.

This tension was particularly visible in Baltimore. The community-generated topic stack identified Johns Hopkins as an important local institution and called for representations that balanced the prestige and employment associated with the institution against its "history of sketchy research

practices." A similar pattern emerged in Tuskegee, where Tuskegee University was closely connected to both community history and contemporary challenges. Workshop materials placed the university within a broader category of "Heritage and Institutional Legacy," which included the Tuskegee Airmen, the Red Tails, the Tuskegee Syphilis Study, George Washington Carver and Booker T. Washington. At the same time, participants identified concerns including "building degradation," "aging facilities," "economic inequality" and inadequate student housing. Another board explicitly identified "Students vs. Locals" as a challenge, demonstrating how the relationship between the university and surrounding community could also involve competition over resources such as housing.

The Baltimore data also demonstrate that institutional relationships extended beyond schools and universities. Participants included Baltimore Gas and Electric (BGE) and the local energy crisis within their "Institutions and Infrastructure" topic stack, identifying customer sentiment about bills, capacity planning and the possible effects of AI and data centers as areas requiring attention. This suggests that institutions become part of community-defined cultural knowledge not only because they represent history or identity, but also because they shape residents' access to resources and their everyday experiences.

4.4 Theme 4: Communities distinguish between acknowledging systemic challenges and being defined by them

Participants incorporated systemic challenges into their descriptions of their communities while also resisting representations that reduced those communities to their problems. Issues such as crime, housing, displacement and lack of resources, were surfaced as important parts of community knowledge without becoming the defining characteristics of the community.

The Tuskegee workshop boards provided a clear example of this distinction. Participants openly identified these systemic challenges including building degradation, aging facilities, lack of public transportation, scarcity of jobs, "economic inequality," limited health-care facilities and inadequate housing. However, participants also identified "nothing to do" as a misconception about Tuskegee and challenged the assumption that the community's small size meant that it lacked "power" or significance. In addition, participants described the community as "one big family" and identified traditions such as line dancing and "Sunday Best," as well as local cultural experiences including lake fishing and neighborhood peace. The boards were therefore used to acknowledge specific community challenges but these challenges did not complete the full description of the community.

Baltimore also reflected a similar distinction, where participants included crime, policing, abandoned housing, food access, transportation and gentrification within discussions of community life, while also identifying a crime-centered image of Baltimore as an incomplete representation of the city. Participants did not shy away from difficult issues, with one participant asking for a report on Baltimore dirt-bike culture that would "Explain the tension between direct bike riders and city officials." However, these concerns existed alongside discussions of local food, music, Black History, schools, sports, festivals and other neighborhood identities. This suggests that the boundary participants drew contextualized representations of systematic challenges and representations, allowing those challenges to stand in for the community as a whole.

The patterns in Baltimore also appeared in Los Angeles, where participants identified housing costs, homelessness, pollution, transportation, fires and gentrification as relevant community concerns. LA participants showed resistance to representations that reduced LA to just Hollywood, celebrity culture and narratives centered around crime and danger. What participants did do was to situate these systematic challenges alongside histories, traditions, institutions, cultural practices and sources of community pride.

5 Discussion

The findings suggest that community-defined cultural knowledge is broader than traditional markers of culture. Across the six REACH communities, participants identified food, music, history and institutions, but they also connected culture to housing, transportation, policing, environmental conditions, local language and the everyday realities of place. These findings also show that cultural understanding depends on local context.

5.1 Social issues as culture: going beyond traditional markers

The findings suggest that participants defined culture as more than food, music, celebrations and traditions. For example, Baltimore participants developed queries about food deserts, literacy rates, Johns Hopkins' relationship with the city and the aftermath of the Freddie Gray uprising, placing structural and institutional issues alongside more conventional cultural topics. These issues are part of the everyday realities that shape what it means to live in Baltimore. In this way, systemic challenges were not separate from cultural knowledge. They provided context for understanding how residents experience their communities, institutions and neighborhoods. This pattern across the six communities suggests that systemic challenges become culturally meaningful when they shape how people experience a place. Housing, transportations, policing, healthcare and environmental conditions may traditionally be understood as social or policy issues, but participants included them into their descriptions of community life. These issues can influence where people

live, how they move through their communities, their relationships with institutions and how particular places change over time. This understanding of culture is consistent with recent calls to rethink how culture is defined in AI research. Alkhamissi et al. (2026) argue that many cultural benchmarks reduce culture to static facts or homogeneous values, even though culture is dynamic, historically situated and experienced through practice. They also identify problems with approaches that treat large geographic or national groups as culturally uniform and overlook differences that exist within communities.

5.2 Cultural knowledge includes local context

The findings also show that cultural knowledge is highly dependent on local context. A broad understanding of a region may identify important facts, but it may not explain why those facts matter to people who live there. Participants frequently connected larger systemic issues to particular neighborhoods, institutions and local histories. In Washington D.C., for example, questions about cultural change were connected to places such as the H Street Corridor and concerns about preserving the city's distinct culture as new residents move into the city. In Taos, questions about healthcare access were connected to Holy Cross Hospital, while concerns about development and outside ownership were connected to Taos Ski Valley.

These examples suggest that understanding a systemic issue requires more than recognizing that the issue exists. It also requires understanding where it is experienced and the meaning it carries within a particular community. This matters for evaluating AI because a response can be factually correct and still miss the local meaning of an issue. The findings suggest that cultural understanding depends not only on identifying a place, institution or problem, but also on recognizing the relationships and histories that make that information meaningful to the community. These findings are particularly important for how cultural understanding is evaluated in generative AI. An AI system may provide factually accurate information about a community while still missing the local context that gives that information meaning. In these exercises, participants challenged assumptions and questioned representations that did not reflect their experiences. This suggests that factual accuracy alone may not be enough to determine whether an AI system demonstrates cultural understanding.

More broadly, the findings demonstrate the importance of involving communities in determining what counts as culturally relevant knowledge. Community members can identify relationships among place, history, institutions and systemic challenges that may not be captured through broader cultural categories or traditional benchmarks. As generative AI increasingly becomes a source of information about people and places, community-defined approaches can provide a more contextual way to evaluate how well these systems understand and represent the communities they describe.

6 Recommendations for Evaluating Model Knowledge of Systemic Issues

Based on the Findings and Discussion, we make the following recommendations when considering model evaluations of hyperlocal cultural knowledge that are inclusive of locally relevant systemic issues and the lived experiences of residents.

6.1 Situate Traditional Cultural Markers within Local Socio-Structural Realities

From the findings in section 4.1, LLM evaluations should not assess traditional cultural markers in isolation (e.g., simply asking for a list of local restaurants or festivals). Instead, evaluation prompts must test whether the model understands how everyday cultural practices are shaped, restricted, or enabled by systemic conditions.

1. **Example**: "How has gentrification and housing displacement in [Neighborhood] affected the historic local art and music scenes?"
2. **Metric**: Grade the model on whether it recognizes the active tension between cultural preservation and structural pressures, rather than providing a sanitized, static list of cultural events.

6.2 Ground Systemic Challenges in Hyperlocal Place-Based Meaning and Vocabulary

As seen in section 4.2, systemic challenges take on unique cultural meaning depending on where they are experienced. Design evaluations should require the model to connect a systemic issue to hyperlocal institutions, geography, and community-specific vocabulary instead of using generic, national, or state-level policy questions.

1. **Example**: "Explain how water access issues shape community identity and history in Northern New Mexico."
2. **Metric**: Assess whether the model can retrieve the localized terminology (acequias) and legal/historical reference points (Abeyta Settlement), rather than outputting generic facts about Southwestern water scarcity.

6.3 Test for Multi-Dimensional Appraisals of Local Institutions

Based on findings in 4.3 Local institutions are primary nodes where history, community pride, and systemic harm collide. It is imperative to create test suites that evaluate whether LLMs provide two-dimensional (purely positive or purely negative) representations of local institutions. Models should be evaluated on their ability to navigate complex, multi-natured institutional relationships.

1. **Example**: "Describe the relationship between Johns Hopkins University and the local communities of Baltimore."
2. **Metric**: Penalize responses that over-index on either extreme (e.g., only discussing the medical school's global prestige or only discussing historical harms). The model must actively balance institutional pride with community-defined systemic tensions.

6.4 Evaluate for "Representational Balance" (Measure Resistance to Deficit-Reductionism)

As shown in section 4.4, participants across all six communities openly discussed intense systemic struggles (crime, policing, medical deserts, unhoused populations) but fiercely resisted representations that reduced their communities to those problems. They demanded that their struggles be situated alongside joy, agency, and community-defined pride—such as Baltimore's dirt bike culture, or Tuskegee's deep familial ties, line dancing, and "Sunday Best." As such, we recommend establishing a "Representational Balance" evaluation metric. This metric should penalize LLMs for two distinct failures:

1. **Superficial Sanitization**: Erasing systemic struggles to present an exoticized, superficial, or tourist-centric caricature (e.g., reducing Los Angeles entirely to Hollywood and celebrity culture).
2. **Deficit-Reductionism**: Presenting a crime-centered or poverty-centered narrative that erases community agency, pride, and daily joy.

An example prompt for this evaluation could be: "Write a descriptive portrait of daily life in Tuskegee, Alabama." In this case the metric could be framed as the model successfully synthesizing local structural challenges (e.g., healthcare access, transit limitations) together with local cultural celebrations and sources of community pride (e.g., lake fishing, neighborhood peace) in a cohesive, non-contradictory narrative.

7 Conclusion

This study examined how systemic challenges are incorporated into community-defined cultural knowledge across six REACH communities. The findings show that participants did not treat issues such as housing, transportation, policing, health care, environmental conditions and economic inequality as separate from culture. Instead, these challenges were connected to everyday experiences, local places, institutions, histories and community identity. At the same time, participants resisted representations that allowed those challenges to define their communities as a whole. This study highlights the importance of including community members in defining culturally relevant knowledge, as social issues may be missed when considering what information is important to surface about local culture for AI systems.

These findings demonstrate that cultural knowledge is not limited to recognizable markers such as food, music, traditions or historical figures. Understanding a community also requires attention to the conditions that shape how people live, move through places, interact with institutions and experience change over time. In this sense, community-defined cultural knowledge is both cultural and structural. It reflects not only what communities value or celebrate, but also the systems and experiences that influence everyday life.

This distinction is especially important for generative AI. The recommendations therefore suggest that evaluating cultural understanding should move beyond factual accuracy and consider whether AI-generated representations reflect relationships among place, history, institutions, structural conditions and lived experience. More broadly, this study demonstrates the value of involving community members in defining what culturally relevant knowledge looks like. As generative AI systems increasingly shape how people access information about communities, community-defined approaches can help ensure that cultural evaluation reflects the complexity of the people and places those systems are attempting to represent.

8 Limitations and Future Work

This study has several limitations that highlight opportunities for future research. First, our qualitative approach prioritizes the depth of participants' lived experiences over statistical generalizability. While these findings offer rich insights into localized cultural perspectives, they are not representative of the broader populations in these six communities. Future studies could employ mixed-methods or quantitative frameworks to evaluate these patterns at scale.

Second, participant recruitment via a third-party vendor introduced variability in cohort sizes, educational backgrounds, occupations, and community tenure. These demographic differences likely influenced the types of cultural knowledge shared. Future work could utilize targeted recruitment to deliberately compare distinct subgroups, such as long-term versus new residents.

Next, the single-day, multi-hour workshop format restricted the time participants had to formulate queries, evaluate AI responses, and deeply reflect. Longitudinal studies spanning multiple sessions—or granting participants continuous, home-based platform access—would allow for more iterative query development and mature evaluations of model accuracy.

Finally, this paper recommended a set of metrics that it did not operationalize. Future work must include demonstrating the recommendations against actual community-written queries and corresponding model responses.

Use of Generative AI

Generative Artificial Intelligence tools were used to assist with checking for overall formatting errors (Latex and document outline) as well as possible grammatical mistakes in the document writing. AI assistance was also used to suggest patterns and trends in the initial stages of the data analysis. The content of this paper: scientific ideas, method design, workshops, data analysis coding, results, and conclusion, are of the authors. The authors reviewed, verified, and edited all AI-assisted outputs.

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